A semi-supervised approach for electricity theft monitoring.

Non-technical losses, mainly electricity theft, impose a severe economic burden on utilities. Most existing studies rely on supervised models that require labeled data, which is costly to obtain in practice.

The dataset from the State Grid Corporation of China covers 1035 days and contains 38,757 normal consumers and 3,615 electricity thieves.

The threshold T is set to the 99.7% percentile of the reconstruction error on the training data.

The LSTM-AE achieved the best performance with an accuracy of 97.31% and an AUC of 0.812, outperforming the Conv-AE and GRU-AE models.

Future work will explore variational autoencoders and multiple adaptive thresholds.